

RS003 Week 8 — English Worksheet

Topic: Variable Speed — Hold Shift to Boost · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week you make the flying **feel good**. The ship's thrusters speed up while you hold **Shift**, and when you let go of the arrow keys it slowly coasts to a stop with **friction**, like drifting through space. You learn about a **velocity** variable that builds up and slows down.

 Words to know: **velocity**, **acceleration**, **friction**, **max speed**, **Shift key**

1 · Predict

Read each piece of code. Before you run it, write what you think will happen.

```
velocity_x = 0
ACCEL = 0.5
velocity_x += ACCEL
ship_x = ship_x + int(velocity_x)
```

velocity_x keeps growing while you hold a key. What happens to the ship's movement — does it move at a steady speed or get faster?


```
if not keys[pygame.K_LEFT] and not keys[pygame.K_RIGHT]:
    velocity_x = velocity_x * FRICTION    # FRICTION = 0.85
```

When you let go of both keys, the velocity is multiplied by 0.85 every loop. What does that do over time?


```
max_v = MAX_SPEED_FAST if keys[pygame.K_LSHIFT] else MAX_SPEED
```

What does holding Shift change about the top speed?

2 · Run It

 **Type the example, run it, and feel the drift**

```

import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
CENTER_X = WIDTH // 2
screen = pygame.display.set_mode((WIDTH, HEIGHT))
clock = pygame.time.Clock()

ship_x = CENTER_X
ship_y = HEIGHT - 100
velocity_x = 0
ACCEL = 0.5
MAX_SPEED = 8
MAX_SPEED_FAST = 16
FRICTION = 0.85

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    keys = pygame.key.get_pressed()
    max_v = MAX_SPEED_FAST if (keys[pygame.K_LSHIFT] or keys[pygame.K_RSHIFT]) else MAX_SPEED

    if keys[pygame.K_LEFT]:
        velocity_x -= ACCEL
    if keys[pygame.K_RIGHT]:
        velocity_x += ACCEL

    if velocity_x > max_v:
        velocity_x = max_v
    if velocity_x < -max_v:
        velocity_x = -max_v

    if not keys[pygame.K_LEFT] and not keys[pygame.K_RIGHT]:
        velocity_x = velocity_x * FRICTION

    ship_x = ship_x + int(velocity_x)

    screen.fill((10, 12, 40))
    pygame.draw.rect(screen, (80, 220, 255), (ship_x - 20, ship_y, 40, 50))
    pygame.draw.polygon(screen, (255, 255, 255), [(ship_x, ship_y - 20), (ship_x - 20, ship_y), (ship_x + 20, ship_y)])

```

```
pygame.display.flip()  
clock.tick(60)  
  
pygame.quit()
```

Hold RIGHT, then let go. Does the ship stop instantly or coast? Now hold Shift and fly. What changes?

3 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Fix it, then explain why the original was wrong.

Bug A — The ship should coast to a stop when you let go, but instead it keeps sliding forever.

```
if keys[pygame.K_LEFT]:
    velocity_x -= ACCEL
if keys[pygame.K_RIGHT]:
    velocity_x += ACCEL
ship_x += int(velocity_x)
```

Hint: nothing slows the velocity down when no key is held.

Write the fixed code (add the friction step):

Why was it wrong? Why does your fix work?

Bug B — The ship is supposed to have a top speed, but holding a key for a long time makes it shoot across the screen far too fast.

```
if keys[pygame.K_RIGHT]:
    velocity_x += ACCEL
ship_x += int(velocity_x)
```

Hint: the velocity is never capped at a maximum.

Write the fixed code (add the speed limit):

Why was it wrong? Why does your fix work?

Bug C — The program crashes because Pygame is given a position with a decimal point.

```
ship_x = ship_x + velocity_x
pygame.draw.rect(screen, CYAN, (ship_x - 20, ship_y, 40, 50))
```

Hint: velocity is a decimal; screen positions must be whole numbers.

Write the fixed code:

Why was it wrong? Why does your fix work?

4 · Modify It

Tune the friction

Try `FRICTION = 0.5`, then `0.85`, then `0.99`. Run each one and let go after flying.

Write how each friction value feels (which one is slippery, which one is grippy):

Make Boost mode look different

Add a visual change while Shift is held — for example, make the thrusters a different colour or bigger.

Describe what you changed and which key state triggers it:

Hint: you already know `keys[pygame.K_LSHIFT]`. Use it to pick a colour the same way the code picks `max_v`.

5 · Make It 📷

 Build a ship with feel

Build a program where:

1. the ship speeds up while held (acceleration),
2. it coasts to a stop when released (friction),
3. Shift gives a faster top speed (boost),
4. it never crashes from decimal positions.

Send a **video** comparing normal flying and Boost flying, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I added a velocity variable that ...

Acceleration makes the ship ...

Friction makes it ... when I let go.

Holding Shift changes the top speed to ...

One tricky moment was when ...

If I had more time, I would ...

6 · Record Your Walkthrough 🎥

Take a video showing normal flying and Boost flying. Talk through it like you are teaching someone. Try to use these words: **velocity**, **acceleration**, **friction**, **max speed**, **Shift**.

1. Fly slowly, then hold Shift and fly fast.
2. Let go of the keys and show the ship coasting.
3. Read your friction line out loud and say what 0.85 does.
4. Show the difference between two friction values you tried.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + your normal/Boost comparison video to teacher on KakaoTalk.